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Cloud Computing Lab

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Section: 5-B

LAB NO 11

https://github.com/23-22411-057-sudo/CC_Saira_2023-BSE-057_Lab11

Task 0 Lab Setup (Codespace & GH CLI)

- taskA_codespace_create_and_list.png

```
PS C:\Users\saira\CC_Saira_2023-BSE-057_Lab11> gh codespace create --repo 23-22411-057-sudo/CC_Saira_2023-BSE-057_Lab11
✓ Codespaces usage for this repository is paid for by 23-22411-057-sudo
? Choose Machine Type: 2 cores, 8 GB RAM, 32 GB storage
vigilant-system-5g7g4x4xw74737p4w
PS C:\Users\saira\CC_Saira_2023-BSE-057_Lab11> gh codespace list
NAME          DISPLAY_NAME    REPOSITORY          BRANCH  STATE    CREATED_AT
studious-fishstick-r454rprpjx... studious fishstick 23-22411-057-sudo/ex... main*   Shutdown about 27 days ago
vigilant-system-5g7g4x4xw7473... vigilant system    23-22411-057-sudo/CC... main    Available less than a minute ago
PS C:\Users\saira\CC_Saira_2023-BSE-057_Lab11>
```

- taskA_codespace_ssh_connected.png

```
PS C:\Users\saira\CC_Saira_2023-BSE-057_Lab11> gh codespace ssh -c vigilant-system-5g7g4x4xw74737p4w
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

Task 1 — Provider & Basic variable (variable precedence)

- task1_touch_main_tf.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ touch main.tf
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ ls
README.md  main.tf
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

- task1_main_tf_provider.png

```

GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

```

- task1_terraform_init.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform -version
Terraform v1.14.3
on linux_amd64
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |

```

- task1_variable_and_output_added.png

```

GNU nano 7.2 main.tf
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

variable "subnet_cidr_block" {
  type      = string
  default   = "10.0.0.0/24"
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

```

- task1_apply_prompt_for_var.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
var.subnet_cidr_block
Enter a value: yes

Changes to Outputs:
+ subnet_cidr_block_output = "yes"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
subnet_cidr_block_output = "yes"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano main.tf

```

- task1_apply_with_default.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
~ subnet_cidr_block_output = "yes" -> "10.0.0.0/24"
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
subnet_cidr_block_output = "10.0.0.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- task1_env_var_set_and_apply.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ export TF_VAR_subnet_cidr_block=10.0.20.0/24
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
~ subnet_cidr_block_output = "10.0.0.0/24" -> "10.0.20.0/24"
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
subnet_cidr_block_output = "10.0.20.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

- task1_terraform_tfvars_and_apply.png

```
GNU nano 7.2                                     terraform.tfvars *
```

```
subnet_cidr_block = "10.0.30.0/24"
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ touch terraform.tfvars
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano terraform.tfvars
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ cat terraform.tfvars
subnet_cidr_block = "10.0.30.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
~ subnet_cidr_block_output = "10.0.20.0/24" -> "10.0.30.0/24"
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
subnet_cidr_block_output = "10.0.30.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- task1_var_override_with_dash_var.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.40.0/24"
```

Changes to Outputs:

```
~ subnet_cidr_block_output = "10.0.30.0/24" -> "10.0.40.0/24"
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
subnet_cidr_block_output = "10.0.40.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- task1_printenv_tf_var_and_unset.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ printenv | grep TF_VAR_
TF_VAR_subnet_cidr_block=10.0.20.0/24
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ unset TF_VAR_subnet_cidr_block
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ printenv | grep TF_VAR_
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

Task 2 — Variable validation & sensitive / ephemeral variables

- task2_subnet_variable_with_validation.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

variable "subnet_cidr_block" {
  type        = string
  default     = ""
  description = "CIDR block to assign to the application subnet"
  sensitive   = false
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[0-9]{1,3}\\.{3}[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
  }
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}
```

- task2_subnet_validation_error.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve -var "subnet_cidr_block=10.0.0"

Error: Invalid value for variable

on main.tf line 6:
  6: variable "subnet_cidr_block" {
    |   var.subnet_cidr_block is "10.0.0"

The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24.

This was checked by the validation rule at main.tf:14,3-13.

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

- task2_api_token_variable_added.png

```
GNU nano 7.2 main.tf *

variable "subnet_cidr_block" {
  type        = string
  default     = ""
  description = "CIDR block to assign to the application subnet"
  sensitive   = false
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[0-9]{1,3}\\.{3}[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
  }
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type        = string
  default     = ""
  description = "Short-lived API session token used during apply operations"
  sensitive   = true
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
  }
}

output "api_session_token_output" {
  value     = var.api_session_token
  sensitive = true
}
```

- task2_api_token_apply_sensitive.png


```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token"

Changes to Outputs:
+ api_session_token_output = (sensitive value)
~ subnet_cidr_block_output = "10.0.40.0/24" -> "10.0.30.0/24"

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

- task2_check_terraform_state_api_token.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ cat terraform.tfstate | grep -C 3 api_session_token_output
"serial": 6,
"lineage": "1212c1f6-39a3-0eaa-0261-77744fd55c65",
"outputs": {
  "api_session_token_output": {
    "value": "my_API_session_Token",
    "type": "string",
    "sensitive": true
  }
}
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- task2_api_token_ephemeral_error.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve -var "api_session_token=my_API_session_Token"

Error: Ephemeral output not allowed

on main.tf line 37:
37: output "api_session_token_output" {

Ephemeral outputs are not allowed in context of a root module

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- task2_api_token_default_apply.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
subnet_cidr_block_output = "10.0.30.0/24"
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

```
GNU nano 7.2                                main.tf *
}

variable "subnet_cidr_block" {
  type        = string
  default     = ""
  description = "CIDR block to assign to the application subnet"
  sensitive   = false
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[0-9]{1,3}\\.[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
    error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
  }
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type        = string
  default     = ""
  description = "Short-lived API session token used during apply operations"
  sensitive   = true
  nullable    = false
  ephemeral   = false

  validation {
    condition     = can(regex("^[A-Za-z0-9-]{20}$", var.api_session_token))
    error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
  }
}

output "api_session_token_output" {
  value     = var.api_session_token
  sensitive = true
}
```

Task 3 — Project-level variables, locals, and outputs

- task3_variables_added.png

```
GNU nano 7.2 main.tf *
default = ""
description = "CIDR block to assign to the application subnet"
sensitive = false
nullable = false
ephemeral = false

validation {
  condition = can(regex("^[0-9]{1,3}\\.{3}[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
  error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type = string
  default = "my_API_session_Token"
  description = "Short-lived API session token used during apply operations"
  sensitive = true
  nullable = false
  ephemeral = false
  validation {
    condition = can(regex("^[A-Za-z0-9-]{20}$", var.api_session_token))
    error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
  }
}

output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}

variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
```

- task3_terraform_tfvars_populated.png

AKIAVHLXGUTLGD7LN2WZ

dqu8S1IW6PfazKT+LDg3A6H3vspE4syQHGY7dSet

```
GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment = "dev"
project_name = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count = 3
monitoring = true
```

- task3_locals_tf_created.png

```
GNU nano 7.2 locals.tf *
locals {
  resource_name = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count = var.subnet_count
  is_production = var.environment == "prod"
  monitoring_enabled = var.monitoring || local.is_production
}
```

- task3_outputs_apply.png

```

GNU nano 7.2                                main.tf *
sensitive = true
nullable = false
ephemeral = false
validation {
  condition = can(regex("[A-Za-z0-9-]{20,}", var.api_session_token))
  error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
}
}
output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
output "resource_name" {
  value = local.resource_name
}

output "primary_public_subnet" {
  value = local.primary_public_subnet
}

output "subnet_count" {
  value = local.subnet_count
}

output "is_production" {
  value = local.is_production
}

output "monitoring_enabled" {
  value = local.monitoring_enabled
}

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab11 (main) \$ terraform apply -auto-approve

```

Changes to Outputs:
+ is_production           = false
+ monitoring_enabled      = true
+ primary_public_subnet   = "subnet-074fbd6f85279d82f"
+ resource_name           = "lab_work-dev"
+ subnet_count            = 3

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count = 3

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab11 (main) \$ |

Task 4 — Maps and Objects

- task4_tags_variable_added.png

```

GNU nano 7.2                                main.tf *
sensitive = true
nullable = false
ephemeral = false
validation {
  condition = can(regex("[A-Za-z0-9-]{20,}", var.api_session_token))
  error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
}
}
output "api_session_token_output" {
  value = var.api_session_token
  sensitive = true
}
variable "environment" {}
variable "project_name" {}
variable "primary_subnet_id" {}
variable "subnet_count" {}
variable "monitoring" {}
output "resource_name" {
  value = local.resource_name
}
variable "tags" {
  type = map(string)
}

output "tags" {
  value = var.tags
}

```

- task4_tags_output.png

```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count      = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}

```

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve

```

Changes to Outputs:

```

+ tags = {
  + Environment = "dev"
  + Owner       = "platform-team"
  + Project     = "sample-app"
}

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
is_production            = false
monitoring_enabled       = true
primary_public_subnet    = "subnet-074fbd6f85279d82f"
resource_name            = "lab_work-dev"
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count             = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})

```

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |

```

- task4_server_config_output.png

```

GNU nano 7.2 main.tf *
ephemeral = false

validation {
  condition = can(regex("^[0-9]{1,3}\\.{3}[0-9]{1,3}/[0-9]+$", var.subnet_cidr_block))
  error_message = "The subnet_cidr_block must be a valid CIDR notation string, such as 10.0.0.0/24."
}

output "subnet_cidr_block_output" {
  value = var.subnet_cidr_block
}

variable "api_session_token" {
  type = string
  default = "my_API_session_Token"
  description = "Short-lived API session token used during apply operations"
  sensitive = true
  nullable = false
  ephemeral = false
  validation {
    condition = can(regex("^[A-Za-z0-9-]{20,}$", var.api_session_token))
    error_message = "The API session token must be at least 20 characters and contain only letters, numbers, hyphens, or underscores."
  }
}

variable "server_config" {
  type = object({
    name = string
    instance_type = string
    monitoring = bool
    storage_gb = number
    backup_enabled = bool
  })
}

output "server_config" {
  value = var.server_config
}

```



```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count     = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

```

Changes to Outputs:
+ server_config = {
+   backup_enabled = false
+   instance_type  = "t3.micro"
+   monitoring     = true
+   name           = "web-server"
+   storage_gb     = 20
+ }

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
is_production            = false
monitoring_enabled      = true
primary_public_subnet    = "subnet-074fbd6f85279d82f"
resource_name            = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type"  = "t3.micro"
  "monitoring"     = true
  "name"           = "web-server"
  "storage_gb"     = 20
}
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count             = 3
tags = tomap({
  "Environment" = "dev"
  "Owner"       = "platform-team"
  "Project"     = "sample-app"
})

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

Task 5 — Collections: list, tuple, set & mutation via locals

- task5_collections_defined.png

```

GNU nano 7.2                                     main.tf *
}
value = local.primary_public_subnet

}

output "subnet_count" {
  value = local.subnet_count
}

output "is_production" {
  value = local.is_production
}

output "monitoring_enabled" {
  value = local.monitoring_enabled
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
}

variable "availability_zones" {
  type = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1b"]
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

```

- task5_compare_collections.png

```

023-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
Changes to Outputs:
+ compare_collections = {
+   list_example = [
+     "web-2",
+     "web-1",
+     "web-2",
+   ]
+   set_example = [
+     "me-central-1a",
+     "me-central-1b",
+   ]
+   tuple_example = [
+     "web-1",
+     4,
+     true,
+   ]
+ }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
023-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $

```

- task5_locals_mutations.png

```
GNU nano 7.2                                locals.tf *
locals {
  resource_name      = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count       = var.subnet_count
  is_production       = var.environment == "prod"
  monitoring_enabled  = var.monitoring || local.is_production
}
locals {
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set = setunion(var.availability_zones, ["me-central-1c"])
}
```

- task5_mutation_comparison.png

```
GNU nano 7.2                                main.tf *
output "primary_public_subnet" {
  value = local.primary_public_subnet
}

output "subnet_count" {
  value = local.subnet_count
}

output "is_production" {
  value = local.is_production
}

output "monitoring_enabled" {
  value = local.monitoring_enabled
}

variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
}

variable "availability_zones" {
  type = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1b"]
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

```
Changes to Outputs:
+ mutation_comparison      = {
+   mutated_tuple = [
+     "4",
+     "true",
+     "web-1",
+     "web-2",
+   ]
+   original_tuple = [
+     "web-1",
+     4,
+     true,
+   ]
+ }

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.
```

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
```

Task 6 — Null, any type & dynamic values

- task6_optional_tag_variable.png

```
GNU nano 7.2 main.tf
value = local.is_production
}

output "monitoring_enabled" {
  value = local.monitoring_enabled
}
variable "server_names" {
  type = list(string)
  default = ["web-2", "web-1", "web-2"]
}

variable "server_metadata" {
  type = tuple([string, number, bool])
  default = ["web-1", 4, true]
}

variable "availability_zones" {
  type = set(string)
  default = ["me-central-1b", "me-central-1a", "me-central-1b"]
}

output "compare_collections" {
  value = {
    list_example = var.server_names
    tuple_example = var.server_metadata
    set_example = var.availability_zones
  }
}

output "mutation_comparison" {
  value = {
    original_tuple = var.server_metadata
    mutated_tuple = local.mutated_tuple
  }
}

variable "optional_tag" {
  type = string
  description = "A tag that may or may not be provided"
  default = null
}
```


- task6_locals_merge.png

```
GNU nano 7.2                                locals.tf *
locals {
  resource_name      = "${var.project_name}-${var.environment}"
  primary_public_subnet = var.primary_subnet_id
  subnet_count       = var.subnet_count
  is_production       = var.environment == "prod"
  monitoring_enabled  = var.monitoring || local.is_production
}
locals {
  mutated_list = setunion(var.server_names, ["web-3"])
  mutated_tuple = setunion(var.server_metadata, ["web-2"])
  mutated_set   = setunion(var.availability_zones, ["me-central-1c"])
}
locals {
  server_tags = merge(
    { Name = "web-server" },
    var.optional_tag != null ? { Custom = var.optional_tag } : {}
  )
}
```

- task6_optional_tag_no_value.png

```
output "optional_tag" {
  value = local.server_tags
}
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
+ optional_tag = {  
  + Name = "web-server"  
}
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>  
compare_collections = {  
  "list_example" = tolist([  
    "web-2",  
    "web-1",  
    "web-2",  
  ])  
  "set_example" = toset([  
    "me-central-1a",  
    "me-central-1b",  
  ])  
  "tuple_example" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
is_production = false  
monitoring_enabled = true  
mutation_comparison = {  
  "mutated_tuple" = toset([  
    "4",  
    "true",  
    "web-1",  
    "web-2",  
  ])  
  "original_tuple" = [  
    "web-1",  
    4,  
    true,  
  ]  
}  
optional_tag = {  
  "Name" = "web-server"  
}  
primary_public_subnet = "subnet-074fbd6f85279d82f"  
resource_name = "lab_work-dev"  
server_config = {  
  "backup_enabled" = false  
  "instance_type" = "t3.micro"  
  "monitoring" = true  
  "name" = "web-server"  
  "storage_gb" = 20  
}
```

- task6_optional_tag_with_value.png

```
GNU nano 7.2 terraform.tfvars *  
subnet_cidr_block = "10.0.30.0/24"  
environment       = "dev"  
project_name      = "lab_work"  
primary_subnet_id = "subnet-074fbd6f85279d82f"  
subnet_count      = 3  
monitoring        = true  
tags = {  
  Environment = "dev"  
  Project     = "sample-app"  
  Owner       = "platform-team"  
}  
server_config = {  
  name           = "web-server"  
  instance_type  = "t3.micro"  
  monitoring     = true  
  storage_gb     = 20  
  backup_enabled = false  
}  
optional_tag = "dev"
```

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```
api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}
subnet_cidr_block_output = "10.0.30.0/24"
subnet_count = 3
tags = tomap({
  "Environment" = "dev"
  "Owner" = "platform-team"
  "Project" = "sample-app"
})
```

- task6_dynamic_value_string.png

```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count     = 3
monitoring       = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}
optional_tag = "dev"
dynamic_value = "hello"

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```
+ value_received = "hello"
```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
server_config = {
  "backup_enabled" = false
  "instance_type" = "t3.micro"
  "monitoring" = true
  "name" = "web-server"
  "storage_gb" = 20
}

```

- task6_dynamic_value_number.png

```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count      = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}
optional_tag = "dev"
dynamic_value = "hello"
dynamic_value = 42

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

```

Changes to Outputs:
  ~ value_received      = "hello" -> 42

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}
primary_public_subnet = "subnet-074fbd6f85279d82f"
resource_name = "lab_work-dev"
server_config = {

```

- task6_dynamic_value_list.png


```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count      = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}
optional_tag = "dev"
dynamic_value = ["a", "b", "c"]

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

```

Changes to Outputs:
+ value_received      = [
+   + "a",
+   + "b",
+   + "c",
+ ]

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}

```

- task6_dynamic_value_map.png

```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count      = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}
optional_tag = "dev"
dynamic_value = {
  name = "server"
  cpu  = 4
}
dynamic_value = {
  name = "server"
  cpu  = 4
}
}

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

```

Changes to Outputs:
  ~ value_received      = 42 -> {
    + cpu = 4
    + name = "server"
  }

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}

```

- task6_dynamic_value_null.png

```

GNU nano 7.2 terraform.tfvars *
subnet_cidr_block = "10.0.30.0/24"
environment       = "dev"
project_name      = "lab_work"
primary_subnet_id = "subnet-074fbd6f85279d82f"
subnet_count      = 3
monitoring        = true
tags = {
  Environment = "dev"
  Project     = "sample-app"
  Owner       = "platform-team"
}
server_config = {
  name           = "web-server"
  instance_type  = "t3.micro"
  monitoring     = true
  storage_gb     = 20
  backup_enabled = false
}
optional_tag = "dev"
dynamic_value = null

```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
```

Changes to Outputs:

```

- value_received = {
  - cpu = 4
  - name = "server"
} -> null

```

You can apply this plan to save these new output values to the Terraform state, without changing any real infrastructure.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

Outputs:

```

api_session_token_output = <sensitive>
compare_collections = {
  "list_example" = tolist([
    "web-2",
    "web-1",
    "web-2",
  ])
  "set_example" = toset([
    "me-central-1a",
    "me-central-1b",
  ])
  "tuple_example" = [
    "web-1",
    4,
    true,
  ]
}
is_production = false
monitoring_enabled = true
mutation_comparison = {
  "mutated_tuple" = toset([
    "4",
    "true",
    "web-1",
    "web-2",
  ])
  "original_tuple" = [
    "web-1",
    4,
    true,
  ]
}
optional_tag = {
  "Custom" = "dev"
  "Name" = "web-server"
}

```

Task 7 — Git ignore

- task7_gitignore_created.png

```

GNU nano 7.2 .gitignore *
.terraform/*
*.tfstate
*.tfstate.*
*.tfvars
*.pem

```

Task 8 — Clean-up then build real infra (VPC, Subnet, IGW, routing, default route table)

- task8_clean_files.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ touch .gitignore
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano .gitignore
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano terraform.tfvars
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ ls terraform.tfvars
terraform.tfvars
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ locals.tf
-bash: locals.tf: command not found
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano locals.tf
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano main.tf
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |
```

- task8_variables_recreated.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
|
```

- task8_vpc_resources_added.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files    = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}
```

- task8_subnet_resources_added.png

```

GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id            = aws_vpc.myapp_vpc.id
  cidr_block        = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

```

- task8_terraform_tfvars_vpc_values.png

```

GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}
variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id            = aws_vpc.myapp_vpc.id
  cidr_block        = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

```

- task8_vpc_subnet_apply.png

```

GNU nano 7.2 terraform.tfvars *
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"

```

- task8_igw_route_table_before_apply.png

```
023-22411-057~sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.27.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
023-22411-057~sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_subnet.myapp_subnet_1 will be created
+ resource "aws_subnet" "myapp_subnet_1" {
  + arn                                = (known after apply)
  + assign_ipv6_address_on_creation    = false
  + availability_zone                  = "me-central-1a"
  + availability_zone_id                = (known after apply)
  + cidr_block                         = "10.0.10.0/24"
  + enable_dns64                       = false
  + enable_resource_name_dns_a_record_on_launch = false
  + enable_resource_name_dns_aaaa_record_on_launch = false
  + id                                 = (known after apply)
  + ipv6_cidr_block_association_id     = (known after apply)
  + ipv6_native                        = false
  + map_public_ip_on_launch            = false
  + owner_id                           = (known after apply)
  + private_dns_hostname_type_on_launch = (known after apply)
  + region                             = "me-central-1"
  + tags                               = {
    + "Name" = "dev-subnet-1"
  }
  + tags_all                             = {
    + "Name" = "dev-subnet-1"
  }
  + vpc_id                               = (known after apply)
}

# aws_vpc.myapp_vpc will be created
+ resource "aws_vpc" "myapp_vpc" {
```

- task8_igw_route_table_after_apply.png

```
GNU nano 7.2 main.tf *
provider "aws" {
  shared_config_files = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id = aws_vpc.myapp_vpc.id
  cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_route_table" "myapp_route_table" {
  vpc_id = aws_vpc.myapp_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}
```

- task8_association_apply.png


```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_internet_gateway.myapp_igw will be created
+ resource "aws_internet_gateway" "myapp_igw" {
+   arn      = (known after apply)
+   id       = (known after apply)
+   owner_id = (known after apply)
+   region   = "me-central-1"
+   tags     = {
+     + "Name" = "dev-igw"
+   }
+   tags_all = {
+     + "Name" = "dev-igw"
+   }
+   vpc_id   = "vpc-0db340e16e7969803"
}

# aws_route_table.myapp_route_table will be created
+ resource "aws_route_table" "myapp_route_table" {
+   arn      = (known after apply)
+   id       = (known after apply)
+   owner_id = (known after apply)
+   propagating_vgws = (known after apply)
+   region   = "me-central-1"
+   route    = [
+     + {
+       + cidr_block      = "0.0.0.0/0"
+       + gateway_id      = (known after apply)
+       + # (11 unchanged attributes hidden)
+     },
+   ]
+   tags     = {
+     + "Name" = "dev-rt"
+   }
+   tags_all = {
+     + "Name" = "dev-rt"
+   }
+   vpc_id   = "vpc-0db340e16e7969803"
}

Plan: 2 to add, 0 to change, 0 to destroy.
aws_internet_gateway.myapp_igw: Creating...
aws_internet_gateway.myapp_igw: Creation complete after 0s [id=igw-0fe85a36f5f1a5415]
aws_route_table.myapp_route_table: Creating...
aws_route_table.myapp_route_table: Creation complete after 1s [id=rtb-075d86430344070a8]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |

```

- task8_default_route_table.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano main.tf
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_route_table.myapp_route_table: Refreshing state... [id=rtb-075d86430344070a8]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# aws_route_table_association.a_rtb_subnet will be created
+ resource "aws_route_table_association" "a_rtb_subnet" {
+   id             = (known after apply)
+   region         = "me-central-1"
+   route_table_id = "rtb-075d86430344070a8"
+   subnet_id      = "subnet-0ec094ba44313f78a"
}

Plan: 1 to add, 0 to change, 0 to destroy.
aws_route_table_association.a_rtb_subnet: Creating...
aws_route_table_association.a_rtb_subnet: Creation complete after 1s [id=rtbassoc-00627ccf08c94f491]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |

```

- task8_default_route_table_apply.png

```

GNU nano 7.2 main.tf
provider "aws" {
  shared_config_files = ["~/aws/config"]
  shared_credentials_files = ["~/aws/credentials"]
}

variable "vpc_cidr_block" {}
variable "subnet_cidr_block" {}
variable "availability_zone" {}
variable "env_prefix" {}
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id = aws_vpc.myapp_vpc.id
  cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

```

```

$23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_route_table_association.a_rtb_subnet: Refreshing state... [id=rtbassoc-00627ccf08c94f491]
aws_route_table.myapp_route_table: Refreshing state... [id=rtb-075d86430344070a8]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create
- destroy

Terraform will perform the following actions:

```

# aws_default_route_table.main_rt will be created
+ resource "aws_default_route_table" "main_rt" {
  + arn = (known after apply)
  + default_route_table_id = "rtb-05454c54ecb86fd35"
  + id = (known after apply)
  + owner_id = (known after apply)
  + region = "me-central-1"
  + route = [
    + {
      + cidr_block = "0.0.0.0/0"
      + gateway_id = "igw-0fe85a36f5f1a5415"
      # (10 unchanged attributes hidden)
    },
  ]
  + tags = {
    + "Name" = "dev-rt"
  }
  + tags_all = {
    + "Name" = "dev-rt"
  }
  + vpc_id = (known after apply)
}

# aws_route_table.myapp_route_table will be destroyed
# (because aws_route_table.myapp_route_table is not in configuration)
- resource "aws_route_table" "myapp_route_table" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-075d86430344070a8" -> null
  - id = "rtb-075d86430344070a8" -> null
  - owner_id = "359416636630" -> null
  - propagating_vgws = [] -> null
  - region = "me-central-1" -> null
  - route = [
    - {
      - cidr_block = "0.0.0.0/0"
      - gateway_id = "igw-0fe85a36f5f1a5415"
      # (11 unchanged attributes hidden)
    },
  ] -> null
  - tags = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id = "vpc-0db340e16e7969803" -> null
}

# aws_route_table_association.a_rtb_subnet will be destroyed
# (because aws_route_table_association.a_rtb_subnet is not in configuration)
- resource "aws_route_table_association" "a_rtb_subnet" {
  - id = "rtbassoc-00627ccf08c94f491" -> null
}

```

Task 9 — Security Group, Key Pair, EC2 Instance, user_data & nginx

- task9_my_ip_variable_added.png

```

GNU nano 7.2 main.tf *
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

resource "aws_subnet" "myapp_subnet_1" {
  vpc_id      = aws_vpc.myapp_vpc.id
  cidr_block  = var.subnet_cidr_block
  availability_zone = var.availability_zone
  tags = {
    Name = "${var.env_prefix}-subnet-1"
  }
}

resource "aws_internet_gateway" "myapp_igw" {
  vpc_id = aws_vpc.myapp_vpc.id
  tags = {
    Name = "${var.env_prefix}-igw"
  }
}

resource "aws_default_route_table" "main_rt" {
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.myapp_igw.id
  }

  tags = {
    Name = "${var.env_prefix}-rt"
  }
}

variable "my_ip" {}
variable "instance_type" {}

```

- task9_public_ip_curl.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ curl icanhazip.com
20.192.21.49
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ nano terraform.tfvars
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ |

```

```

GNU nano 7.2 terraform.tfvars *
vpc_cidr_block      = "10.0.0.0/16"
subnet_cidr_block   = "10.0.10.0/24"
availability_zone    = "me-central-1a"
env_prefix           = "dev"
my_ip                = "20.192.21.49/32"
instance_type        = "t3.micro"

```

- task9_security_group_apply.png

```
GNU nano 7.2 main.tf *
}

tags = {
  Name = "${var.env_prefix}-rt"
}
}
variable "my_ip" {}
variable "instance_type" {}
resource "aws_default_security_group" "myapp_sg" {
  vpc_id = aws_vpc.myapp_vpc.id

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "${var.env_prefix}-sg"
  }
}
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-05454c54ecb86fd35]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# aws_default_security_group.myapp_sg will be created
+ resource "aws_default_security_group" "myapp_sg" {
+   arn                = (known after apply)
+   description        = (known after apply)
+   egress              = [
+     {
+       cidr_blocks     = [
+         "0.0.0.0/0",
+       ]
+       from_port        = 0
+       ipv6_cidr_blocks = []
+       prefix_list_ids  = []
+       protocol         = "-1"
+       security_groups  = []
+       self              = false
+       to_port          = 0
+     }
+   ],
+   id                  = (known after apply)
+   ingress             = [
+     {
+       cidr_blocks     = [
+         "0.0.0.0/0",
+       ]
+       from_port        = 80
+       ipv6_cidr_blocks = []
+       prefix_list_ids  = []
+       protocol         = "tcp"
+       security_groups  = []
+       self              = false
+       to_port          = 80
+     }
+   ],
+   {
+     cidr_blocks     = [
+       "20.192.21.49/32",
+     ]
+     from_port        = 22
+   }
+ }
```

```

+ prefix_list_ids = []
+ protocol        = "tcp"
+ security_groups = []
+ self            = false
+ to_port         = 80
+ # (1 unchanged attribute hidden)
},
+ {
+   cidr_blocks = [
+     "20.192.21.49/32",
+   ]
+   from_port   = 22
+   ipv6_cidr_blocks = []
+   prefix_list_ids = []
+   protocol     = "tcp"
+   security_groups = []
+   self         = false
+   to_port      = 22
+   # (1 unchanged attribute hidden)
+ },
]
+ name = (known after apply)
+ name_prefix = (known after apply)
+ owner_id = (known after apply)
+ region = "me-central-1"
+ revoke_rules_on_delete = false
+ tags = {
+   + "Name" = "dev-sg"
+ }
+ tags_all = {
+   + "Name" = "dev-sg"
+ }
+ vpc_id = "vpc-0db340e16e7969803"
}

```

Plan: 1 to add, 0 to change, 0 to destroy.

aws_default_security_group.myapp_sg: Creating...

aws_default_security_group.myapp_sg: Creation complete after 2s [id=sg-04474168d96055b72]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) \$

- task9_keypair_created_and_saved.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ aws ec2 create-key-pair \
--key-name MyED25519Key \
--key-type ed25519 \
--key-format pem \
--query 'KeyMaterial' \
--output text > MyED25519Key.pem

```

chmod 600 MyED25519Key.pem

ls -l MyED25519Key.pem

-rw----- 1 codespace codespace 388 Jan 2 17:06 MyED25519Key.pem

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) \$ |

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ cat .gitignore

```

.terraform/*

*.tfstate

.tfstate.

*.tfvars

*.pem

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) \$ |

- task9_instance_type_set.png


```

GNU nano 7.2                                     main.tf *
resource "aws_default_security_group" "myapp_sg" {
  vpc_id = aws_vpc.myapp_vpc.id

  ingress {
    from_port = 22
    to_port   = 22
    protocol  = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    from_port = 80
    to_port   = 80
    protocol  = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    from_port = 0
    to_port   = 0
    protocol  = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }

  tags = {
    Name = "${var.env_prefix}-sg"
  }
}

resource "aws_instance" "myapp_server" {
  ami           = "ami-05524d6658fcf35b6"
  instance_type = var.instance_type
  subnet_id     = aws_subnet.myapp_subnet_1.id
  vpc_security_group_ids = [aws_default_security_group.myapp_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name        = "MyED25519Key"

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}

output "aws_instance_public_ip" {
  value = aws_instance.myapp_server.public_ip
}

```

- task9_ec2_apply_and_public_ip.png

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-04474168d96055b72]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-05454c54ecb86fd35]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```

# aws_instance.myapp_server will be created
+ resource "aws_instance" "myapp_server" {
  + ami                     = "ami-05524d6658fcf35b6"
  + arn                    = (known after apply)
  + associate_public_ip_address = true
  + availability_zone       = "me-central-1a"
  + disable_api_stop       = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized           = (known after apply)
  + enable_primary_ipv6     = (known after apply)
  + force_destroy           = false
  + get_password_data       = false
  + host_id                 = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile    = (known after apply)
  + id                     = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle      = (known after apply)
  + instance_state          = (known after apply)
  + instance_type           = "t3.micro"
  + ipv6_address_count      = (known after apply)
  + ipv6_addresses          = (known after apply)
  + key_name                = "MyED25519Key"
  + monitoring              = (known after apply)
  + outpost_arn             = (known after apply)
  + password_data           = (known after apply)
  + placement_group         = (known after apply)
  + placement_group_id      = (known after apply)
  + placement_partition_number = (known after apply)
  + primary_network_interface_id = (known after apply)
  + private_dns             = (known after apply)
  + private_ip              = (known after apply)
  + public_dns              = (known after apply)
  + public_ip               = (known after apply)
  + region                  = "me-central-1"
  + secondary_private_ips   = (known after apply)
  + security_groups         = (known after apply)
  + source_dest_check       = true

```



```

GNU nano 7.2                                     main.tf *
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
}

egress {
    from_port    = 0
    to_port      = 0
    protocol     = "-1"
    cidr_blocks  = ["0.0.0.0/0"]
}

tags = {
    Name = "${var.env_prefix}-sg"
}
}

resource "aws_instance" "myapp_server" {
    ami                = "ami-05524d6658fcf35b6"
    instance_type      = var.instance_type
    subnet_id          = aws_subnet.myapp_subnet_1.id
    availability_zone   = var.availability_zone
    associate_public_ip_address = true
    key_name            = aws_key_pair.ssh_key.key_name

    tags = {
        Name = "${var.env_prefix}-ec2-instance"
    }
}

output "aws_instance_public_ip" {
    value = aws_instance.myapp_server.public_ip
}

resource "aws_key_pair" "ssh_key" {
    key_name    = "serverkey"
    public_key = file("~/ssh/id_ed25519.pub")
}

```

```

@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform apply -auto-approve
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-04474168d96055b72]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-05454c54ecb86fd35]
aws_instance.myapp_server: Refreshing state... [id=i-04b91120c9ba2328]

```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

```

+ create
-/+ destroy and then create replacement

```

Terraform will perform the following actions:

```

# aws_instance.myapp_server must be replaced
-/+ resource "aws_instance" "myapp_server" {
    ~ arn                = "arn:aws:ec2:me-central-1:359416636630:instance/i-04b91120c9ba2328" -> (known after apply)
    ~ disable_api_stop   = false -> (known after apply)
    ~ disable_api_termination = false -> (known after apply)
    ~ ebs_optimized      = false -> (known after apply)
    + enable_primary_ipv6 = (known after apply)
    - hibernation         = false -> null
    + host_id             = (known after apply)
    + host_resource_group_arn = (known after apply)
    + iam_instance_profile = (known after apply)
    ~ id                  = "i-04b91120c9ba2328" -> (known after apply)
    ~ instance_initiated_shutdown_behavior = "stop" -> (known after apply)
    + instance_lifecycle = (known after apply)
    ~ instance_state      = "running" -> (known after apply)
    ~ ipv6_address_count  = 0 -> (known after apply)
    ~ ipv6_addresses      = [] -> (known after apply)
    ~ key_name            = "MyED25519Key" -> "serverkey" # forces replacement
    ~ monitoring          = false -> (known after apply)
    + outpost_arn         = (known after apply)
    + password_data       = (known after apply)
    + placement_group     = (known after apply)
    + placement_group_id  = (known after apply)
    ~ placement_partition_number = 0 -> (known after apply)
    ~ primary_network_interface_id = "eni-0d76057335509d2d2" -> (known after apply)
    ~ private_dns         = "ip-10-0-10-140.me-central-1.compute.internal" -> (known after apply)
    ~ private_ip          = "10.0.10.140" -> (known after apply)
    + public_dns          = (known after apply)
    ~ public_ip           = "158.252.72.213" -> (known after apply)
    ~ secondary_private_ips = [] -> (known after apply)
    ~ security_groups     = [] -> (known after apply)
}

```

Plan: 2 to add, 0 to change, 1 to destroy.

Changes to Outputs:

```
aws_instance_public_ip = "158.252.72.213" -> (known after apply)
aws_instance.myapp_server: Destroying... [id=i-04b911120c9ba2328]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 00m10s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 00m20s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 00m30s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 00m40s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 00m50s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 01m00s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-04b911120c9ba2328, 01m10s elapsed]
aws_instance.myapp_server: Destruction complete after 1m10s
aws_key_pair.ssh_key: Creating...
aws_key_pair.ssh_key: Creation complete after 0s [id=serverkey]
aws_instance.myapp_server: Creating...
aws_instance.myapp_server: Still creating... [00m10s elapsed]
aws_instance.myapp_server: Creation complete after 13s [id=i-08f48c3787f540b0a]
```

```
Apply complete! Resources: 2 added, 0 changed, 1 destroyed.
```

Outputs:

```
aws_instance_public_ip = "3.28.207.191"
```

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

```
[ec2-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main)] $ ssh ec2-user@3.28.207.191
The authenticity of host '3.28.207.191 (3.28.207.191)' can't be established.
ED25519 key fingerprint is SHA256:3AX5lF9leC7MZozavLwvhwKAUs0asX90WOXF+XLABLU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.28.207.191' (ED25519) to the list of known hosts.
```

```
#_
~\##### Amazon Linux 2023
NN\#####
NN\###|
NN\#/ https://aws.amazon.com/linux/amazon-linux-2023
NNV~!'->
    NNN
      NN.-.
        _/_/_/_
          _/m/'-$
```

- task9 nginx local curl.png

```
[ec2-user@ip-10-0-10-214 ~]$ sudo systemctl start nginx
[ec2-user@ip-10-0-10-214 ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-10-0-10-214 ~]$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[ec2-user@ip-10-0-10-214 ~]$
```

- task9 nginx browser page.png

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

```
023-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057-Lab11 (main) $ terraform destroy -auto-approve
aws_key_pair.ssh_key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-04474168d96055b72]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-05454c54ecb86fd35]
aws_instance.myapp_server: Refreshing state... [id=i-08f48c3787f540b0a]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
- destroy

Terraform will perform the following actions:

# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-05454c54ecb86fd35" -> null
  - default_route_table_id = "rtb-05454c54ecb86fd35" -> null
  - id = "rtb-05454c54ecb86fd35" -> null
  - owner_id = "359416636630" -> null
  - propagating_vgws = [] -> null
  - region = "me-central-1" -> null
  - route
    - {
      - cidr_block = "0.0.0.0/0"
      - gateway_id = "igw-0fe85a36f5f1a5415"
      # (10 unchanged attributes hidden)
    }
  ] -> null
  - tags = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id = "vpc-0db340e16e7969803" -> null
}

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:security-group/sg-04474168d96055b72" -> null
  - description = "default VPC security group" -> null
  - egress
    - {
      - cidr_blocks = [
        - "0.0.0.0/0",
      ]
      - from_port = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids = []
      - protocol = "-1"
      - security_groups = []
      - self = false
      - to_port = 0
      # (1 unchanged attribute hidden)
    }
  ],
}
```

Cleanup

- cleanup_destroy.png


```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ terraform destroy -auto-approve
```

```
aws_key_pair.ssh_key: Refreshing state... [id=serverkey]
aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0db340e16e7969803]
aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-0fe85a36f5f1a5415]
aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-0ec094ba44313f78a]
aws_default_security_group.myapp_sg: Refreshing state... [id=sg-04474168d96055b72]
aws_default_route_table.main_rt: Refreshing state... [id=rtb-05454c54ecb86fd35]
aws_instance.myapp_server: Refreshing state... [id=i-08f48c3787f540b0a]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- destroy

Terraform will perform the following actions:

```
# aws_default_route_table.main_rt will be destroyed
- resource "aws_default_route_table" "main_rt" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:route-table/rtb-05454c54ecb86fd35" -> null
  - default_route_table_id = "rtb-05454c54ecb86fd35" -> null
  - id = "rtb-05454c54ecb86fd35" -> null
  - owner_id = "359416636630" -> null
  - propagating_vgws = [] -> null
  - region = "me-central-1" -> null
  - route = [
    - {
      - cidr_block = "0.0.0.0/0"
      - gateway_id = "igw-0fe85a36f5f1a5415"
      # (10 unchanged attributes hidden)
    },
  ] -> null
  - tags = {
    - "Name" = "dev-rt"
  } -> null
  - tags_all = {
    - "Name" = "dev-rt"
  } -> null
  - vpc_id = "vpc-0db340e16e7969803" -> null
}

# aws_default_security_group.myapp_sg will be destroyed
- resource "aws_default_security_group" "myapp_sg" {
  - arn = "arn:aws:ec2:me-central-1:359416636630:security-group/sg-04474168d96055b72" -> null
  - description = "default VPC security group" -> null
  - egress = [
    - {
      - cidr_blocks = [
        - "0.0.0.0/0",
      ]
      - from_port = 0
      - ipv6_cidr_blocks = []
      - prefix_list_ids = []
      - protocol = "-1"
      - security_groups = []
      - self = false
      - to_port = 0
      # (1 unchanged attribute hidden)
    },
  ],
}
```

```
- arn = "arn:aws:ec2:me-central-1:359416636630:vpc/vpc-0db340e16e7969803" -> null
- assign_generated_ipv6_cidr_block = false -> null
- cidr_block = "10.0.0.0/16" -> null
- default_network_acl_id = "acl-0e402f097acfa07a0" -> null
- default_route_table_id = "rtb-05454c54ecb86fd35" -> null
- default_security_group_id = "sg-04474168d96055b72" -> null
- dhcp_options_id = "dopt-07a746a94dc2af666" -> null
- enable_dns_hostnames = false -> null
- enable_dns_support = true -> null
- enable_network_address_usage_metrics = false -> null
- id = "vpc-0db340e16e7969803" -> null
- instance_tenancy = "default" -> null
- ipv6_netmask_length = 0 -> null
- main_route_table_id = "rtb-05454c54ecb86fd35" -> null
- owner_id = "359416636630" -> null
- region = "me-central-1" -> null
- tags = {
  - "Name" = "dev-vpc"
} -> null
- tags_all = {
  - "Name" = "dev-vpc"
} -> null
# (4 unchanged attributes hidden)
}
```

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:

```
- aws_instance.public_ip = "3.28.207.191" -> null
aws_default_security_group.myapp_sg: Destroying... [id=sg-04474168d96055b72]
aws_default_route_table.main_rt: Destroying... [id=rtb-05454c54ecb86fd35]
aws_instance.myapp_server: Destroying... [id=i-08f48c3787f540b0a]
aws_default_security_group.myapp_sg: Destruction complete after 0s
aws_default_route_table.main_rt: Destruction complete after 0s
aws_internet_gateway.myapp_igw: Destroying... [id=igw-0fe85a36f5f1a5415]
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 00m10s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0fe85a36f5f1a5415, 00m10s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 00m20s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0fe85a36f5f1a5415, 00m20s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 00m30s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0fe85a36f5f1a5415, 00m30s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 00m40s elapsed]
aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0fe85a36f5f1a5415, 00m40s elapsed]
aws_internet_gateway.myapp_igw: Destruction complete after 47s
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 00m50s elapsed]
aws_instance.myapp_server: Still destroying... [id=i-08f48c3787f540b0a, 01m00s elapsed]
aws_instance.myapp_server: Destruction complete after 1m0s
aws_key_pair.ssh_key: Destroying... [id=serverkey]
aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0ec094ba44313f78a]
aws_key_pair.ssh_key: Destruction complete after 1s
aws_subnet.myapp_subnet_1: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0db340e16e7969803]
aws_vpc.myapp_vpc: Destruction complete after 1s
```

Destroy complete! Resources: 7 destroyed.

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```

- cleanup_state_files.png

- cleanup_verify_no_secrets.png

```
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        .terraform.lock.hcl
        aws/
        awscliv2.zip
        id_ed25519.pub
        locals.tf
        main.tf
        main.tf.save

nothing added to commit but untracked files present (use "git add" to track)
@23-22411-057-sudo → /workspaces/CC_Saira_2023-BSE-057_Lab11 (main) $
```